A formal proof of the graph-normalization adjacency inequality

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25 October 2026

Abstract

We prove the graph-normalization adjacency inequality for every finite simple graph and every strictly positive weight vector. Connectivity is unnecessary. The proof is a weighted sum of the elementary inequality $2ab \le a^2 + b^2$ and applies to any symmetric nonnegative matrix whose action on the initial vector is strictly positive. Lean verifies the entire matrix statement. This is a formal verification of a standard majorization consequence; no priority claim is made.

1 Statement

Let A be the adjacency matrix of a finite simple undirected graph, put $B = A + I$, and let $x_i > 0$ for every vertex. Set

$$s_i = (Bx)_i, \qquad y_i = \frac{x_i}{s_i}.$$

The diagonal entries of B are one, so $s_i \ge x_i > 0$.

Theorem 1. *For every such graph and vector,*

$$y^T(A + I)y \le \|y\|_1.$$

This is Conjecture 8.1, equation (19), of Guigues [1], the target of MathDB 349070. The source assumes the graph connected; the theorem above removes that assumption. The conclusion is solely this inequality, not convergence of the normalization iteration.

2 A symmetric-matrix inequality

Lemma 1. *Let B be a finite symmetric matrix with nonnegative entries, and let $x_i \ge 0$ and $a_i \in \mathbb{R}$. Writing $s_i = \sum_j B_{ij}x_j$, one has*

$$\sum_{i,j} B_{ij}(x_i a_i)(x_j a_j) \le \sum_i x_i a_i^2 s_i.$$

Proof. Multiply $2a_ia_j \le a_i^2 + a_j^2$ by $B_{ij}x_ix_j \ge 0$ and sum over all ordered pairs. Symmetry and exchange of the finite summation indices give

$$\begin{aligned} 2\sum_{i,j} B_{ij}x_ix_ja_ia_j &\le \sum_{i,j} B_{ij}x_ix_ja_i^2 + \sum_{i,j} B_{ij}x_ix_ja_j^2 \\ &= 2\sum_i x_ia_i^2 \sum_j B_{ij}x_j. \end{aligned}$$

Dividing by two proves the lemma. □

Proof of Theorem 1. More generally, suppose B satisfies Lemma 1 and all $s_i > 0$. Substitute $a_i = 1/s_i$. Then $x_i a_i = y_i$ and $x_i a_i^2 s_i = y_i$, so the lemma gives $y^T B y \le \sum_i y_i$. For a graph, $B = A + I$ satisfies all hypotheses. Since $y_i > 0$, its sum is exactly $\|y\|_1$. □

3 Formal correspondence and prior status

The Lean source is `formal/GraphNormalization.lean`. Its general finite-sum lemma is `symmetric_energy_le`; the substitution is `normalized_energy_le`. The explicit graph kernel equals mathlib's adjacency matrix plus the identity by `closedKernel_eq_adj_add_one`. The theorem `mathdb349070` states the result using mathlib matrix-vector multiplication and dot product, with the ℓ^1 norm written as the sum of absolute values. Positivity of every denominator and the absolute-value equality are proved internally.

The diagonal majorization bound is standard and appears explicitly in Appendix B of Guigues's later paper [2]. Its displayed global-dominance inequality, with $y^k = x$ and test vector $y_i = x_i/(Bx)_i$, already yields the conclusion above. Our contribution is the direct elementary argument and its complete formal certificate; no new-to-literature assertion is made. The later paper's additional convergence claims are neither used nor verified here.

The release contains the independent review, dependency pins, and `verify_outcome.py`. Run the verifier on the extracted package with Lean 4.34.1 and the pinned mathlib dependencies. All eight local definitions and theorems are checked, with only `propext`, `Classical.choice`, and `Quot.sound` permitted. The formal proof covers arbitrary finite graphs, including disconnected graphs; it does not use numerical testing or an assumed matrix inequality.

References

[1] Laurent Guigues. *Concerning Iterative Graph Normalization and Maximum Weight Independent Sets*. arXiv:2012.07764v1, 2020. Conjecture 8.1, equation (19). https://arxiv.org/html/2012.07764v1.

[2] Laurent Guigues. *Graph Normalization: Fast Binarizing Dynamics for Differentiable MWIS*. arXiv:2605.05330v1, 2026. Appendix B, global dominance. https://arxiv.org/html/2605.05330v1.